Machinery noise in industry

Hearing damage from occupational noise is one of the most frequently reported occupational diseases in Denmark. The course “Machinery noise in industry” was about understanding where the noise of machines comes from, how it spreads in a factory hall, and how it can be controlled – preferably at the source. The page has an interactive factory hall and a figure on vibration isolation.

> DTH, course 8044 (1988), 6 points: Machinery noise in industry, Acoustics Laboratory. The subject is closely linked to the thesis: noise carried through ventilation ducts, and how it is reduced with absorbent linings in the ducts.

Where does the noise come from?

Almost all machinery noise can be traced to three types of source:

Mechanical noise

Impacts and vibrations in solid parts: gears, bearings, punching, riveting, imbalance in rotating parts. The forces set plates and casings vibrating, and they radiate sound. The noise often contains tones at the rotation speed and its harmonics – in a gearbox at the tooth-meshing frequency = number of teeth × revolutions per second.

Aerodynamic noise

Turbulence and vortices in flowing air: fans, compressed-air exhausts, jets and valves. Lighthill showed in 1952 that the sound power of a free jet grows as the eighth power of the flow velocity – halve the exhaust velocity and the noise falls by 24 dB. Fans have a tone at the blade-passing frequency.

Hydraulic noise

Pressure pulsations in liquids: pumps, cavitation and flow in valves. The pulsations travel through the pipework and make pipes and tanks radiate sound far from the pump.

Source strength

A source that is small compared with the wavelength radiates inefficiently: a monopole radiates with $W \propto \omega^2$, a dipole (a small vibrating plate, a force) with ω^4, and a quadrupole (turbulence) with ω^6. That is why it is so hard to get bass out of a small loudspeaker.

Sound power – the fingerprint of the source

The sound pressure at a machine depends on the room, the distance and other machines. The sound power L_W depends only on the machine itself. Since the 1980s manufacturers in the EU have had to declare the noise of many machines as a sound power level, and there are three standardised ways to determine it:

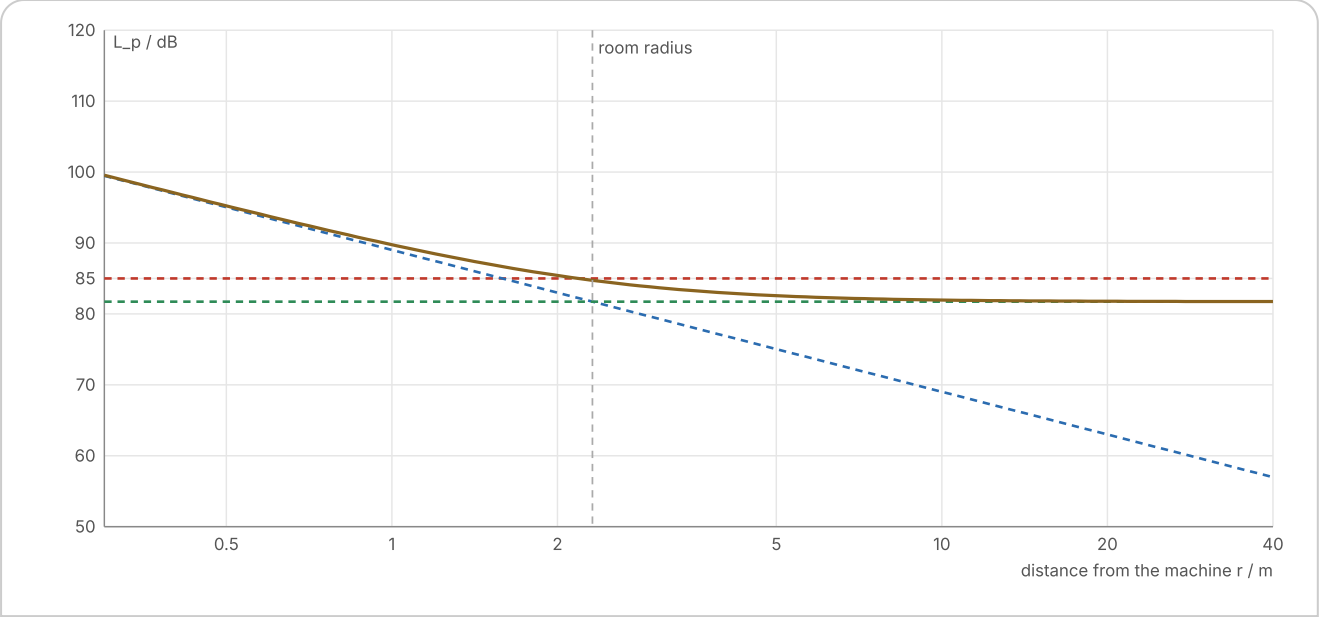
- **Free field over a reflecting plane** (ISO 3744/3745): the sound pressure is measured at a number of points on a surface around the machine, and the power passing through the surface is calculated: $L_W = \overline{L_p} + 10\log_{10}(S/1\ \mathrm{m^2})$.
- **Reverberation room** (ISO 3741): in a room with hard walls the sound field is diffuse, and the sound pressure is proportional to the power – if the absorption of the room is known from the reverberation time, the power is known.
- **Sound intensity** (ISO 9614): a surface around the machine is scanned with a two-microphone intensity probe. The method also works in a noisy factory, because noise from other sources entering and leaving the surface cancels out.

Noise in a factory hall

Indoors one hears both the direct sound from the machine and sound reflected many times from walls, ceiling and floor. The reflected field is roughly equally strong everywhere in the room (a diffuse field), and its strength is set by the absorption of the room. Sabine's theory gives

$$L_p = L_W + 10\log_{10}\left(\frac{Q}{4\pi r^2} + \frac{4}{R}\right), \qquad R = \frac{S\bar{\alpha}}{1-\bar{\alpha}}$$

where S is the total surface area of the room and $\bar{\alpha}$ the mean absorption coefficient. Close to the machine the direct field dominates; further away the diffuse field. The distance at which the two are equal is the *room radius* (reverberation radius) $r_h = \sqrt{QR/16\pi}$. In a typical factory hall it is only a couple of metres. That means a worker standing 10 m from the machine mainly hears reflected sound – which can be reduced with absorbing ceilings. The person tending the machine stands in the direct field, and for him the ceiling hardly helps.



Noise control – source, path and receiver

The basic strategy is to intervene as early as possible in the chain source → transmission path → receiver:

Where	Examples	Typical effect
At the source	Slower fans with more blades, helical gears, balancing, silencers on compressed-air exhausts, rubber-lined conveyors	5–20 dB, often cheapest
Structure-borne sound	Vibration isolation, damping layers on plates, soft pipe mountings and bellows	5–20 dB
Airborne sound in the room	Enclosures, screens, absorbing ceilings, silencers in ducts	Enclosure 10–30 dB, absorption 3–10 dB
At the receiver	Sound-insulated control room, shorter exposure, hearing protectors	Hearing protectors 10–30 dB in practice

Hearing protectors are the last resort, not the first: they are often worn incorrectly or not all the time, and if they are removed for just half an hour of a working day they lose much of their effect, because the dose is counted in energy.

Enclosures and barriers

A box around the machine only works if it is heavy, airtight and lined inside. The insulation of the wall itself follows the mass law (see room acoustics), but inside a closed, hard box the sound builds up through repeated reflections, so the insertion loss of the enclosure becomes roughly

$$D \approx R_{\text{wall}} + 10 \log_{10} \bar{\alpha}_{\text{inside}}$$

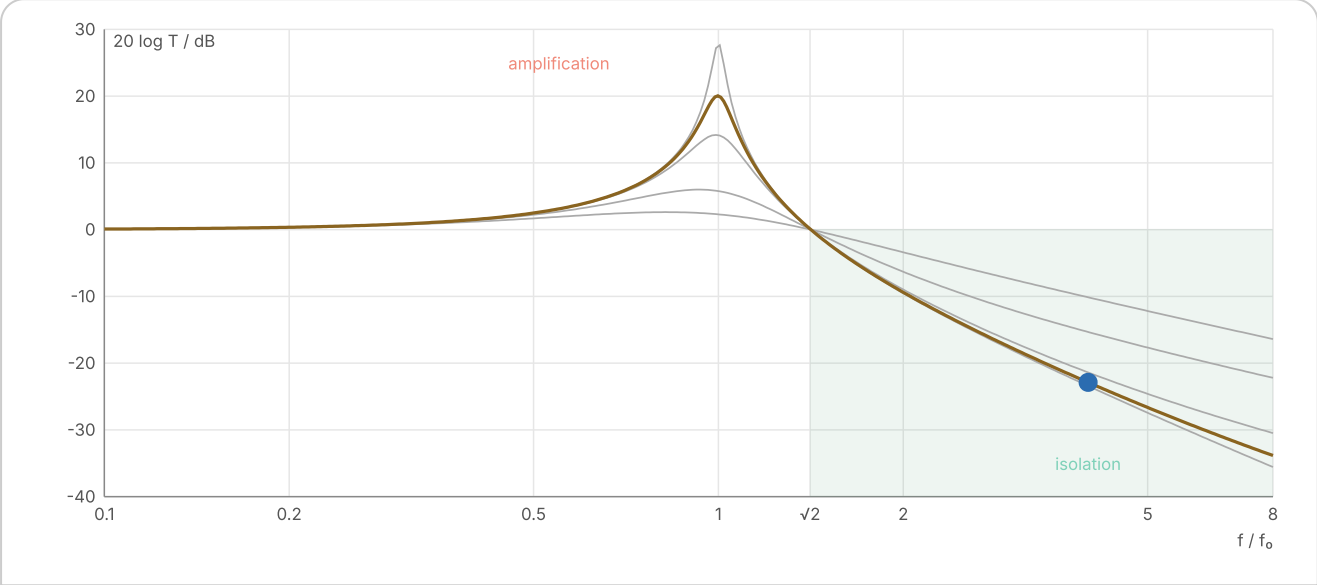
With $\bar{\alpha} = 0.05$ inside the box 13 dB of the wall's insulation is lost; with 50 mm mineral wool ($\bar{\alpha} \approx 0.7$) only 1–2 dB. And an opening of just 1 % of the surface limits the insertion loss to about 20 dB, however thick the wall – sound leaks through holes like water. Openings for ventilation and materials must therefore have silencers.

Vibration isolation

A machine standing directly on a concrete floor sets the whole building vibrating, and the noise then comes out everywhere. If the machine is put on springs (rubber, steel springs, air bellows), the system gets a natural frequency $f_0 = \frac{1}{2\pi}\sqrt{k/m}$. The force transmitted to the floor, relative to the force acting on the machine, is the transmissibility

$$T = \sqrt{\frac{1 + (2\zeta\, f/f_0)^2}{\left(1 - (f/f_0)^2\right)^2 + (2\zeta\, f/f_0)^2}}$$

where ζ is the damping ratio. Isolation only begins when $f > \sqrt{2}\, f_0$; around f_0 the springs amplify the vibrations. The springs must therefore be soft enough to put the natural frequency well below the machine's lowest operating frequency. A practical rule: the static deflection is $\delta = g/(2\pi f_0)^2$ – a spring that sags 25 mm under the machine's weight gives $f_0 \approx 3$ Hz.



Noise and the working environment

The EU noise directive (2003/10/EC), implemented in Danish law, is based on the equivalent noise exposure over an 8-hour working day, $L_{EX,8h}$, and the peak value $L_{C,peak}$:

	$L_{EX,8h}$	$L_{C,peak}$	Requirement
Lower action value	80 dB(A)	135 dB(C)	Information, hearing protectors made available
Upper action value	85 dB(A)	137 dB(C)	Noise-control programme, hearing protectors must be worn
Exposure limit (with protectors)	87 dB(A)	140 dB(C)	Must not be exceeded

With the 3 dB rule, 85 dB(A) for 8 hours corresponds to 100 dB(A) for a quarter of an hour. Prolonged noise above 80–85 dB(A) causes permanent hearing loss, first appearing as a “notch” in the audiogram around 4 kHz, and often tinnitus.

Since 1989

- **The Machinery Directive** (1989, latest 2023/1230) requires manufacturers to design machines for low noise and to state the sound power level.
- **Variable-speed drives:** Fans and pumps are now controlled by variable speed instead of dampers and valves – saving both energy and many decibels.
- **Condition monitoring:** Vibration and noise analysis is used to detect worn bearings and gears before they fail – from simple measurements to machine learning on data from many sensors.
- **Hearing damage:** The number of reported hearing injuries in Denmark has fallen since the 1990s, but they remain among the most common occupational diseases.

Further reading

- István L. Vér and Leo L. Beranek (eds.): *Noise and Vibration Control Engineering* (2nd ed., Wiley 2006)
- David A. Bies, Colin H. Hansen and Carl Q. Howard: *Engineering Noise Control* (5th ed., CRC 2017)
- Danish Working Environment Authority: executive order and guidance on noise at work (in Danish)
- On this site: The silencer in 3D · Lined duct model